

1. The table gives the initial monthly premiums for males and females aged 50-54 as a function of the amount of life insurance. Complete parts a to d.

Life Insurance Amount	Male Premium	Female Premium
\$20,000	\$11.32	\$5.13
30,000	14.15	6.42
50,000	28.29	12.83
75,000	42.44	19.25
100,000	56.58	25.67

- a. If the function $f(x)$ is the premium for males for life insurance amount x , in dollars, find $f(100,000)$ and $f(30,000)$ in dollars.

$f(100,000) =$ (Type an integer or a decimal.)

$f(30,000) =$ (Type an integer or a decimal.)

- b. If the function $g(x)$ is the premium for females for life insurance amount x , in dollars, find $g(50,000)$ and $g(100,000)$ in dollars.

$g(50,000) =$ (Type an integer or a decimal.)

$g(100,000) =$ (Type an integer or a decimal.)

- c. For what value of x is $f(x) = 14.15$?

$x =$ (Type an integer or a decimal.)

- d. Is $f(x)$ greater than $g(x)$ for all x -values?

☐ Yes

☐ No

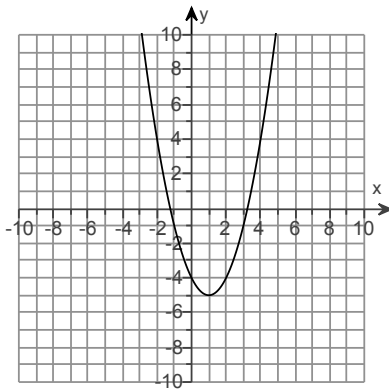
2. Watch the video and then solve the problem given below.

[Click here to watch the video.](#)¹

For each of the functions $y = f(x)$ described below, find $f(-1)$.

(a) $y = 11 - 2x^2$

(b)

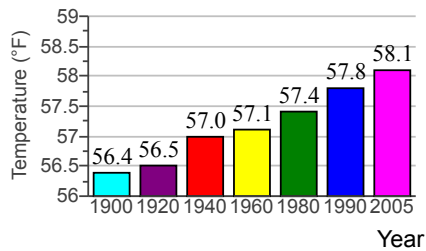


(a) $f(-1) =$

(b) $f(-1) =$

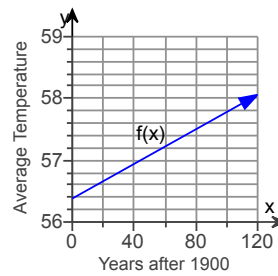
1: <https://mediaplayer.pearsoncmg.com/assets/CA5Hfy010105>

- *3. The bar graph shows the average temperature for a certain city, in degrees Fahrenheit, for seven selected years.



The data can be modeled by the linear function

$f(x) = 0.014x + 56.38$, where $f(x)$ is the average temperature, in degrees Fahrenheit, x years after 1900. The graph of f is shown to the right. Complete parts (a) through (b).



- a. Find and interpret $f(60)$. Use the equation for $f(x)$ to identify this information as a point on the graph.

The point is .
(Type an ordered pair.)

Interpret this information.

The average temperature in the city in the year was °F.

- b. Does $f(60)$ overestimate or underestimate the actual data shown by the bar graph? By how much?

The value $f(60)$ (1) _____ the data in the bar graph by °F.

- (1) ☐ overestimates
☐ underestimates